

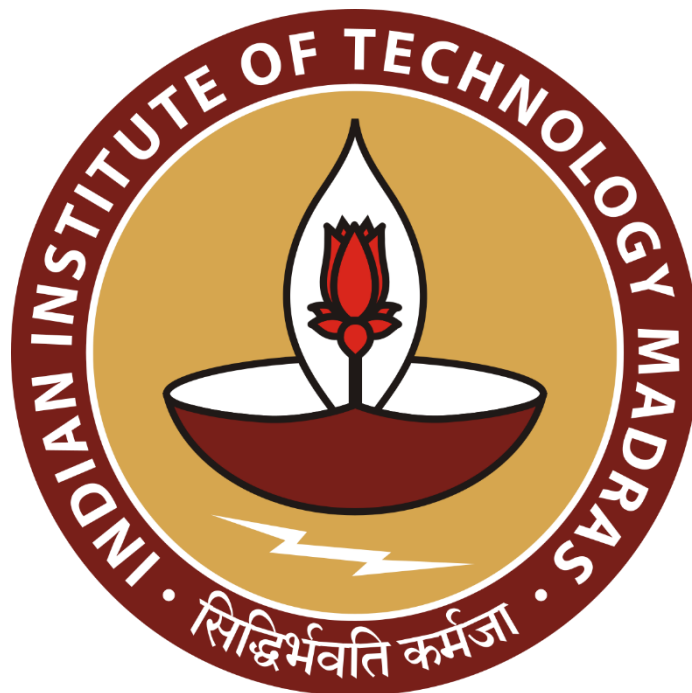
Data-Driven Strategies for Optimizing Profitability and Operations in a Saree Weaving Business

A Midterm Report for the BDM Capstone Project

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1. Executive Summary

Vishwas Maurya Textile, a small-scale saree weaving business located in Chhoti Koiran, Lohta, Varanasi, specializes in producing designer sarees for B2B retailers. The organization faces key operational challenges such as inadequate tracking of production costs, limited understanding of design and color-wise sales forecasting, and the absence of production efficiency metrics. These challenges have resulted in low profitability and irregular workflow due to inconsistent machine utilization and unrecorded operational downtime. The project aims to use data-driven methods to understand these inefficiencies and develop insights for improving production management and profitability.

This midterm report is based on primary data collected directly from business through multiple interactions, telephonic discussions with the business owner and from handwritten transaction records maintained by Vishwas Maurya Textile. The records were digitized and organized into Microsoft Excel for analysis. The dataset comprises sales data of designer sarees (Lachha Buti and Aara designs) from April to October 2025 and purchase data of raw materials such as Taani, Bana, and Jari. Additionally, color-wise sales data were also included, which is only data maintain from the commencement of project. The descriptive statistics highlight variations in sales volume, purchase cost, and color preferences, forming the foundation for identifying cost patterns and forecasting demand.

The analysis so far involves data cleaning, descriptive statistics, and visualization in Excel to identify sales trends, production gaps, and material cost variations. Through graphical representation and trendline analysis, it was observed that Lachha Buti sarees maintain more consistent sales performance compared to Aara designs, indicating their higher market demand and customer preference. Whereas, Aara designs show greater fluctuations in both quantity and price, suggesting issues such as irregular production or changing consumer interest. Analysis of cost side, the purchase data of raw materials revealed that Jari contributes the most to the total production cost, followed by Bana and Taani, which directly influences the profit margin of premium sarees. Visual trends from color-wise sales data also show that bright colors such as red and green sell faster, which useful for future production planning. These findings will support the next phase of the project, which focuses on tracking cost efficiency, optimizing raw material procurement, and applying forecasting models to predict high-demand saree designs and colors, enabling data-driven decision-making for improved productivity and profitability.

2. Proof of Data Originality

To establish the authenticity of the data, the supporting evidence as listed below:

1. Letter from the Organization: Access to the letter can be obtained through the G-Drive link:

[<https://drive.google.com/file/d/1wWQc0ZgkdW9ahQuwAt5y7HnG228Ds6xn/view?usp=sharing>]

2. Interview with Owner : They can be accessed through the G-Drive link:

[https://drive.google.com/drive/folders/11nCCyLfVvQ7w_jwMcZGGH4uHCG-lCd?usp=sharing]

3. Images of Organization: They can be accessed through the G-Drive link:

[<https://drive.google.com/drive/folders/1yBn3sNVz6-OKmXSJUnzrQmbhcu9BdriH?usp=sharing>]

4. Link to Sheet for data collected:

[<https://docs.google.com/spreadsheets/d/1XHX5R9-1G8kp5PguiQSpEBaXbe3-OwUP/edit?usp=sharing&oid=104234663566505786452&rtpof=true&sd=true>]

3.METADATA AND DESCRIPTIVE STATISTICS

3.1 METADATA

The data was collected through multiple interaction, telephonic conversation with the owner and as well as handwritten records book maintained by Vishwas Maurya Textile. The book was unstructured and containing mix of various transactions related to business such as sales of saree and purchase of raw materials.

Gathered data was loaded into MS Excel, containing datasets:

- Sales Data – Three separate Excel sheets from April to October 2025.
- Purchase Data – A single consolidated sheet covering the same period.

The above dataset well maintained in such a way to easily understand relevant details and to combine & connect each excel sheets for further analysis, the following metadata can be defined:

Dataset Title: Sales Data of Lachha Buti Design Saree

Data Collection Period: 12th April 2025 – 29th October 2025

Data Source: Manually recorded transaction data from the saree business production and sales log.

Data Frequency: Weekly or bi-weekly entries (based on production or sales cycle).

Unit of Observation: Each row represents one recorded transaction or batch production entry of Lachha Buti Design Saree.

Variable Name	Format	Data Type	Description / Definition	Importance
Date	DD-MM-YYYY	Date	Represents the date on which the saree sales transaction occurred.	Helps in time-based trend analysis, seasonal demand and identifying peak or low sales periods.
Quantity of Lachha Buti Design Saree	Numeric	Integer (Count)	Number of sarees sold on that date.	Indicates product demand. Useful for inventory control, demand forecasting, and identifying best-selling periods.
Value (₹)	Numeric	Integer (INR Currency)	Total sales amount generated on that specific date.	Helps to measure revenue per transaction.
Price per Quantity (₹)	Numeric	Float (INR Currency)	Average price per saree, calculated as $\text{Value} \div \text{Quantity}$.	Reflects unit selling price, which is key to tracking pricing strategies and profitability margins.

Relevance to Problem Statement

1. The variables Value and Price per Quantity enable calculation of revenue per saree and comparison with production cost to identify profit margins.
2. The Date and Quantity fields help build time-series models to forecast future sales for different designs or color patterns.
3. The Quantity over time (Date) helps evaluate how consistently the production meets demand and how output efficiency varies weekly or monthly.

Dataset Title: Sales Data of Aara Design Saree

Data Collection Period: 12th April 2025 – 29th October 2025

Data Source: Manually recorded data from the saree business production and sales ledger.

Data Frequency: Weekly or near-weekly entries, depending on production/sales activity.

Unit of Observation: Each row represents a production or sale record of Aara Design Sarees for a given date.

Variable Name	Format	Data Type	Description / Definition	Importance
Date	DD-MM-YYYY	Date	The date on which Aara Design sarees were sold.	Helps identify trends and seasonality in sales, allowing for planning of stock.
Quantity of Aara Design Saree	Numeric	Integer (Count)	Number of sarees sold on that date.	Helps in to tracks product demand and assisting in production and sales strategy planning.
Price per Quantity (₹)	Numeric	Float (INR Currency)	Selling price per saree unit.	Shows the pricing pattern and helps in revenue calculation

Relevance to Problem Statement

1. The Price per Quantity field enables calculation of revenue per saree. Combined with production cost estimates, it allows profit margin analysis per batch.
2. The Date and Quantity variables enable creation of time-series models to forecast upcoming sales demand for the Aara Design saree.
3. The Quantity over Date trend helps identify irregular or zero-output days (like 07-06-2025, 13-06-2025, 20-06-2025), signaling efficiency gaps or production delays.

Dataset Title: Colour-wise Sales Data of Saree

Data Collection Period: 21st April 2025 – 29th October 2025

Data Source: Manually recorded colour-wise sales and production data from saree inventory and production logs.

Data Frequency: Weekly or near-weekly entries, depending on colour production and sales batches.

Unit of Observation: Each row represents total saree production/sales for one day, divided by colour categories.

Variable Name	Format	Data Type	Description	Importance
Date	DD-MM-YYYY	Date	The date on which saree sales occurred.	Enables chronological tracking of colour demand and trend forecasting.
Red Color	Numeric	Integer (Count)	Number of red sarees sold on that date.	Helps identify popularity and demand of red sarees.
Dark Green Color	Numeric	Integer (Count)	Number of dark green sarees sold.	Measures seasonal or customer preference for dark green sarees.
Dark Purple Color	Numeric	Integer (Count)	Number of dark purple sarees sold on that date.	Tracks premium or low-demand colour categories.

Bright Purple Color	Numeric	Integer (Count)	Number of bright purple sarees sold.	Helps in analyzing the appeal of vibrant color variants among customers.
Bright Pink Color	Numeric	Integer (Count)	Number of bright pink sarees sold on that date.	Tracks sales trends for popular and fast-selling colour shades.
Sky Blue Color	Numeric	Integer (Count)	Number of sky-blue sarees sold.	Captures trends for lighter-coloured designs.
Deep Red Color	Numeric	Integer (Count)	Number of deep red sarees sold.	Differentiates from standard red sarees; helps track sub-tone preferences.
Dark Blue Color	Numeric	Integer (Count)	Number of dark blue sarees sold on that date.	Represents cooler, formal colors — valuable for assessing diversity in design preferences.
Total	Numeric	Integer (Count)	The total number of sarees sold across all colors on that specific date.	Provides an overall measure of daily sales performance. Used for comparing sales volume with color distribution and business trends.

Relevance to Problem Statement

1. Colour-level production tracking allows cost-per-colour computation since dyeing and design costs vary. This helps identify high-cost or low-profit colour segments.
2. This dataset directly addresses this gap. By analyzing Date-wise and Colour-wise production, one can forecast demand for specific colours using moving averages or regression models.
3. Total column shows output variation across dates. Sudden drops indicate inefficiencies or raw material issues. Colour distribution also reflects production focus and balance across batches.

3.2 DESCRIPTIVE STATISTICS

here is a summarized overview using descriptive statistics. The key insights have been condensed to emphasize the most relevant descriptive statistics for both sales and inventory data:-

Quantity of Lachha Buti Design Saree		Value		Price per Quantity	
Mean	7.485714286	Mean	2678.571429	Mean	350.4908663
Standard Error	0.758560588	Standard Error	293.2381207	Standard Error	13.8719995
Median	6	Median	2100	Median	366.6666667
Mode	4	Mode	1500	Mode	400
Standard Deviation	4.487704959	Standard Deviation	1734.820118	Standard Deviation	82.06785578
Sample Variance	20.1394958	Sample Variance	3009600.84	Sample Variance	6735.132952
Kurtosis	0.661139885	Kurtosis	-0.997076308	Kurtosis	0.539550781
Skewness	0.786663097	Skewness	0.5370581	Skewness	-0.506918345
Range	20	Range	5800	Range	333.3333333
Minimum	1	Minimum	200	Minimum	166.6666667
Maximum	21	Maximum	6000	Maximum	500
Sum	262	Sum	93750	Sum	12267.18032
Count	35	Count	35	Count	35

Figure 3.2.1: Value analysis of Lachha Buti Design Saree sales data

The descriptive statistics of the Lachha Buti Design Saree sales data (Figure 3.2.1) show that, on average, around 7 to 8 sarees are sold in each transaction, though most sales are around 6 sarees, and sometimes only 4 sarees, which is the most common quantity. This means the number of sarees sold usually stays small but sometimes increases due to bulk or festive orders. The average total value per sale is about ₹2,678, but it changes a lot from one sale to another — sometimes as low as ₹200 and as high as ₹6,000. This big difference happens because some customers buy in large quantities, which increases the total amount. On the other hand, the price per saree stays quite stable, mostly between ₹350 to ₹400, showing that the shop follows a consistent pricing policy. Overall, the data tells us that while the price per saree does not change much, the total sales value and variation in quantity sold. The sales data is positively skewed, meaning there are a few transactions with very high quantities and high values. There are no extreme outliers, which means the business runs in a steady and predictable way. In simple words, the saree prices remain almost fixed, but the total income depends on how many sarees customers buy in one order.

The descriptive statistics of Lachha Buti Design Sarees (Figure 3.2.1) reveal variations in sales volume, average selling price, and customer preferences. These insights help identify which designs are in higher demand and guide stock management and pricing strategies to improve profitability.

Quantity of Aara Design Saree		Price per Quantity		Value	
Mean	3.4	Mean	613.57143	Mean	2086.4286
Standard Error	0.304586784	Standard Error	2.1358429	Standard Error	186.10431
Median	3	Median	625	Median	1875
Mode	2	Mode	625	Mode	1250
Standard Deviation	1.801959718	Standard Deviation	12.635817	Standard Deviation	1101.0079
Sample Variance	3.247058824	Sample Variance	159.66387	Sample Variance	1212218.5
Kurtosis	1.798754618	Kurtosis	-2.090685	Kurtosis	1.4981489
Skewness	1.339670136	Skewness	-0.179864	Skewness	1.2827257
Range	8	Range	25	Range	4800
Minimum	1	Minimum	600	Minimum	600
Maximum	9	Maximum	625	Maximum	5400
Sum	119	Sum	21475	Sum	73025
Count	35	Count	35	Count	35

Figure 3.2.2: Value analysis of Aara Design Sarees sales data

The sales data of Aara Design Sarees (Figure 3.2.2) shows that on average, about 3 to 4 sarees were sold per period, with a mean quantity of 3.4 and a median of 3, which means sales are usually around that number. The standard deviation (1.80) indicates that the sales quantity slightly varies around the average, showing moderate consistency in demand. The skewness value (1.33) suggests that a few higher sales pushed the average upward, meaning some weeks had unusually high sales. The kurtosis (1.79) tells us the data is a bit more peaked than normal, meaning most sales are close to the average.

The price per quantity is quite stable, with an average of ₹613.57, and both median and mode at ₹625, showing that this is the most common price point. The small standard deviation (12.63) means prices didn't change much, staying nearly the same across time. In terms of total value, the average sales value is ₹2086.43, with some large variations (standard deviation = 1101.00), showing that while prices were stable, total earnings fluctuated depending on the number

of sarees sold. The positive skewness (1.28) shows that there were a few high-revenue periods that pulled the average up. Overall, the Aara Design Saree (Figure 3.2.2) shows stable pricing with moderate but consistent demand, and a few weeks of strong sales boosted the total revenue. The sales data of Aara Design Sarees highlight trends in demand, revenue contribution, and seasonal performance. By analyzing these patterns, businesses can optimize inventory and focus on top-performing designs to enhance sales efficiency.

Taani Quantity		Taani Cost		Bana Quantity		Bana Cost		Jari Quantity		Jari Cost	
Mean	5.937964286	Mean	126.25	Mean	6.19696429	Mean	112.5	Mean	8.498214	Mean	60
Standard Error	1.689137304	Standard Error	35.50525801	Standard Error	1.43737824	Standard Error	20.15564437	Standard Error	2.318129	Standard Error	18.25742
Median	0	Median	0	Median	3.28	Median	210	Median	0	Median	0
Mode	0	Mode	0	Mode	0	Mode	210	Mode	0	Mode	0
Standard Deviation	8.938074474	Standard Deviation	187.8761659	Standard Deviation	7.60589074	Standard Deviation	106.653645	Standard Deviation	12.26638	Standard Deviation	96.60918
Sample Variance	79.88917529	Sample Variance	35297.4537	Sample Variance	57.8495739	Sample Variance	11375	Sample Variance	150.4642	Sample Variance	9333.333
Kurtosis	-1.02593498	Kurtosis	-1.32287842	Kurtosis	0.46696715	Kurtosis	-2.13528994	Kurtosis	-1.116967	Kurtosis	-1.075846
Skewness	0.928719607	Skewness	0.847641561	Skewness	1.10918228	Skewness	-0.1514608	Skewness	0.88816	Skewness	1.003249
Range	23.96	Range	445	Range	26.595	Range	210	Range	29.63	Range	210
Minimum	0	Minimum	0	Minimum	0	Minimum	0	Minimum	0	Minimum	0
Maximum	23.96	Maximum	445	Maximum	26.595	Maximum	210	Maximum	29.63	Maximum	210
Sum	166.263	Sum	3535	Sum	173.515	Sum	3150	Sum	237.95	Sum	1680
Count	28	Count	28	Count	28	Count	28	Count	28	Count	28

Figure 3.2.3: Value analysis of purchase data of Taani, Bana, and Jari materials

The purchase data of Taani, Bana, and Jari materials (Figure 3.2.3) shows interesting patterns about their quantity and cost variation. On average, the Taani Quantity purchased was around 5.94 units, while its mean cost was ₹126.25, with a large standard deviation (187.87) indicating big fluctuations in spending. The positive skewness (0.92) suggests that most Taani purchases were small, but a few large orders increased the average. The wide range (23.96 in quantity) confirms irregular buying behavior, possibly due to demand-based or season.

For Bana, the average quantity was 6.19 units, with a mean cost of ₹112.5. The standard deviation (7.65 for quantity and 106.65 for cost) indicates some variability, but less than Taani. Its positive skewness (1.10) again shows a few large purchases influenced the overall mean. Meanwhile, Jari Quantity averaged 8.49 units, higher than both Taani and Bana, and its mean cost was ₹60, the lowest among all three materials. Although the variation (standard deviation of 12.26 for quantity) shows inconsistent purchases, the positive skewness (0.88) suggests small but frequent buys with occasional large ones. Overall, these statistics shows that Jari was purchased more frequently but at lower cost, Bana was moderately consistent, and Taani purchases were the most variable in both cost and quantity may be due to differences in design requirements or production schedules. The purchase data analysis of Taani, Bana, and Jari materials (Figure 3.2.3) shows material consumption and cost patterns. Understanding these helps control expenses, forecast material requirements, and ensure consistent production quality.

	Red Color	Dark Green Color	Dark Purple Color	Bright Purple Color	Bright Pink Color	Sky Blue Color	Deep Red Color	Dark Blue Color
count	50	50	50	50	50	50	50	50
mean	2.08	0	0.1	0.88	2.76	0.98	0.34	0.12
min	0	0	0	0	0	0	0	0
25%	0	0	0	0	2	0	0	0
50%	2	0	0	0	2	0.5	0	0
75%	3.75	0	0	2	4	2	0	0
max	7	0	2	4	8	4	3	2
std	1.736	0	0.416	1.189	1.901	1.116	0.772	0.435

Figure 3.2.4: Value analysis of color-wise sales data of sarees data

The color-wise sales data of sarees (Figure 3.2.4) shows that among all available shades, Bright Pink and Red colors are the most popular among customers. The Bright Pink Color has the highest mean (2.76), indicating it was sold the most frequently, followed by Red Color (mean 2.08) and Sky-Blue Color (mean 0.98). This suggests that customers prefer vibrant and bright colors, especially pink and red tones, while darker shades like Dark Green, Dark Purple, and Dark Blue show very low or zero sales, reflecting weak demand.

The standard deviation values show how much variation exists in the sales of each color. Bright Pink (std 1.90) and Red (std 1.73) have relatively higher variation, meaning their sales were not consistent but had periods of high demand. On the other hand, colors like Dark Purple, Deep Red, and Dark Blue have very low standard deviation values (below 1), indicating steady but minimal sales. Overall, this data clearly shows that customers are more attracted to bright, lively saree colors like pink, red, and sky blue, whereas darker shades are less favored, helping the business identify which color varieties should be prioritized for better sales performance. Color-wise sales statistics reveal which colors are most preferred by customers. These insights assist in designing collections that match customer tastes and improving production planning for better market response.

4. DETAILED EXPLANATION OF ANALYSIS PROCESS AND METHOD

4.1 DATA CLEANING AND PREPROCESSING

The process of data analysis encompasses different components such as defining the problem statements, data collection, data organizing, data cleaning, applying analysis techniques, and drawing conclusions. The analysis process for the project also involves a combination of quantitative and qualitative methods, each one chosen for their ability to address specific aspects of the business challenges.

The first step is most challenging part of this journey was data acquisition, due to tourist place and crowded place in most part of Varanasi city, most of business owner doesn't record business data whom 'so ever I engaging and some of businesses rejections due fear of privacy, some relative friends willing to share data verbally, but hesitant about sharing any raw data.

Finally, after searching couple of months a gracious business owner was willing to share the data, establishing one of the most important milestones of data acquisition.

The raw data collected from Vishwas Maurya Textile was entirely maintained in rough handwritten formats.

Since no digital records existed, the next stage of preprocessing involved manually transcribing every transaction into Microsoft Excel. This process was time-consuming but ensured high data accuracy and minimized the need for extensive cleaning later. Once digitized, the dataset was checked for missing entries, incorrect dates, duplicate rows, and inconsistent units.

Columns were standardized to uniform formats, such as converting all quantities to numerical values, ensuring consistent use of colour names, and verifying purchase and sales entries against the business owner's notes. Outliers were identified through visual inspection and descriptive statistics. However, since many fluctuations genuinely represented market or production variations, only clearly erroneous values—such as negative quantities or incomplete dates—were corrected.

Additional preprocessing included separating composite entries into structured fields (e.g., saree design, cost, quantity, colour) and creating time-based variables to facilitate trend and forecasting analysis. Through this systematic cleaning and transformation process, the final dataset became reliable, structured, and ready for statistical and exploratory analysis.

The saree sales dataset required careful cleaning and preprocessing before any meaningful analysis could be performed. Since the data came from multiple manually recorded entries—covering saree sales and purchase details—it was essential to bring everything into a consistent structure.

Key preprocessing steps carried out such as:

Naming columns for clarity (e.g., Quantity of Aara Design Saree, Price per Quantity, Color-wise Sales, Taani/Bana/Jari Purchase Data). Sorted datasets chronologically to prepare for time-series analysis. Normalized uneven decimal places in Taani/Bana/Jari quantities. Replaced blanks with zero only where logically correct (e.g., colour not sold on that date).

4.2 METHODS/ANALYSIS

After preprocessing, the cleaned dataset allowed for detailed insights by Extensively utilizing spreadsheets and their functions for various calculations essential to the analysis process. Spreadsheets provided a familiar and user-friendly interface for conducting complex calculations and aggregating data functions such as SUM, AVERAGE, etc.

The analysis for this project employed both quantitative and qualitative methods to understand the operational, cost, and sales dynamics of the business. Descriptive statistics formed the foundation of the quantitative analysis, using Excel to compute measures such as mean, standard deviation, variance, skewness, kurtosis, and range for raw material costs and saree sales.

These metrics helped identify variability in purchase prices, sales performance across designs, and patterns within the color-wise demand. Time-series analysis was applied to the month-wise sales and purchase data to uncover seasonal trends, production gaps, and changes in customer preferences.

Visualization tools such as line charts, bar graphs, and trendlines (created using Excel) were used to interpret these patterns and compare design-wise performance between Lachha Buti and Aara sarees. The qualitative

component included insights gained from multiple conversations with the business owner, which helped contextualize the numerical findings—particularly regarding machine downtime, challenges in material procurement, and reasons behind irregular production cycles. Together, these analytical methods allowed the project to identify key inefficiencies, highlight cost drivers, and lay the groundwork for forecasting models and strategic recommendations in the later phases of the study.

5. RESULTS AND FINDINGS

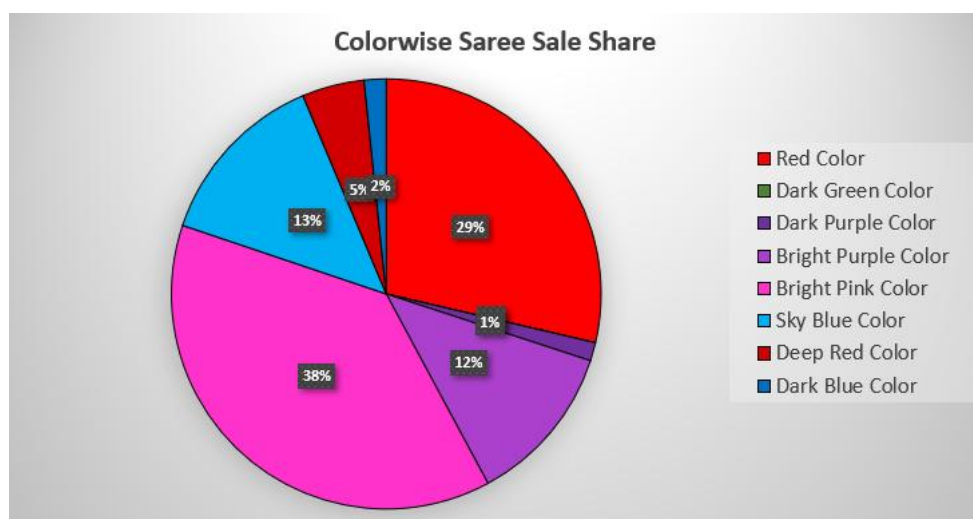


Figure 5.1 Colorwise Saree Sale Share

The color-wise pie chart offers clear results about customer color preferences for designer sarees during the April–October 2025 period. The visualization shows that Bright Pink contributes the largest share of sales at 38%, making it the most preferred color among customers. It's indicating that bright, vibrant shades strongly influence buying decisions. The second highest share is Red Color at 29%, reflecting consistent customer preference for culturally significant colors commonly associated with festivals, weddings, and ceremonies. Following these dominant colors, Sky Blue (13%) and Dark Purple (12%) show moderate contributions, indicating they hold stable but not substantial market share. Were as Deep Red (5%), Dark Blue (2%), and Bright Purple (1%) — have minimal sales shares, suggesting limited demand for these shades. The results clearly highlight that the product mix of the business must align with these high-performing colors to maximize sales.

The color-wise sales distribution demonstrates highly responsive to bold, vibrant sarees that align with cultural preferences in North India and traditionally in demand for festivals, weddings, and special occasions.

These two colors alone constitute 67% of total sales—indicating that a large portion of revenue is concentrated in a limited product range. This also means that the business has a clear opportunity to optimize production by prioritizing these shades. Conversely, colors like Bright Purple (1%) and Dark Blue (2%) contribute negligible sales, indicating that their production may be inefficient and possibly causing wastage of raw materials. This finding highlights the need for targeted color-based production planning, where high-demand colors receive priority and low-demand colors are either scaled down or discontinued.



Figure 5.2 Total Sale

The total sales line chart reveals critical insights regarding daily and seasonal fluctuations. The early period from late April to mid-May shows consistent sales between 12 to 16 sarees per day, pointing toward stable demand and smooth operational flow. However, from early June, the sales sharply decline to values between 1 and 6 sarees, marking the first major downturn in the dataset. This phase shows scattered and unpredictable sales activity, indicating underlying operational inefficiencies or external seasonal factors.

As the timeline progresses, several spikes appear most notably, the 21 sarees sold on 18 August, which is the highest recorded sale in the entire period. Similarly, late October also records high sales of 19 and 15 sarees, suggesting improved sales near festive periods. Overall, the visualization demonstrates a cycle of highs and lows, portraying that demand is irregular and heavily influenced by external market dynamics, seasonality, or production constraints.

The sharp decline in sales during June and July suggests possible production disruptions, machine downtime, maybe due to high temperature in summer thereafter months also align with monsoon season, which may impact both production and customer buying behavior. The coupled with sudden spikes—such as the 21 sarees sold on 18 August—indicates that the business is not fully prepared to capitalize on surges in demand.

The inconsistent pattern also reflects the absence of forecasting tools or data-driven inventory management. Another finding is the improvement in sales during late October, correlating with festival season (Navratri, Diwali), showing that the business performs significantly better during cultural high-demand periods. These findings underscore the need for seasonal production planning, buffer inventory, and better prediction models to leverage peak periods and stabilize low periods.

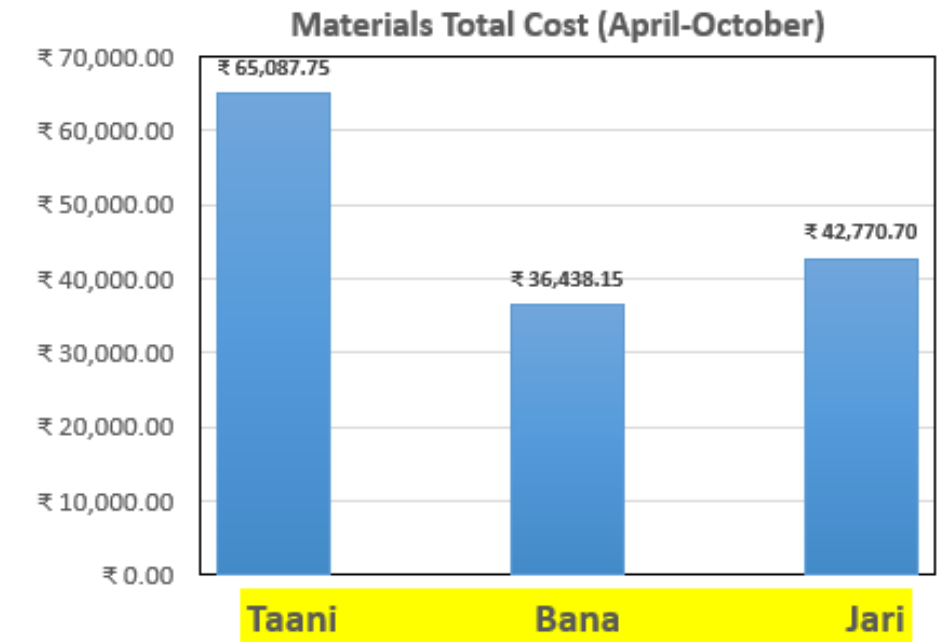


Figure 5.3 Materials Total Cost

The bar chart presents the total cost of key raw materials—Taani, Bana, and Jari—from April to October. Taani accounts for the highest total cost at ₹65,087.75, confirming its central role in production. This could be due to higher consumption rates or frequent purchases required to meet ongoing production needs. Jari, although traditionally an expensive material, stands second with a cost of ₹42,770.70, indicating its significant but variable contribution to designer sarees.

Bana, which is a supportive yarn, records a cost of ₹36,438.15, making it the least expensive but still essential component. The results clearly establish that raw material costs are substantial and heavily influence profitability. Jari and Taani, in particular, represent major cost drivers and require strategic procurement planning. High variability in Taani cost shows potential price fluctuations, inconsistent purchase quantities, or inefficient vendor management.

The finding that Jari and Bana together contribute nearly equal cost also indicates that cost management strategies must consider all materials—not just premium ones. An important observation is that material cost is disproportionately high relative to sales fluctuations, meaning cost control mechanisms are critical for ensuring profitability. These findings suggest the business must improve vendor negotiation, bulk purchasing, and budgeting practices to stabilize operational expenses.